

Social Landscape Analyzer: A Location-Based Social Media Trend Analytics System

A IDP report submitted by

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Under the Guidance of

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CERTIFICATE

This is to certify that the project entitled "**Social Landscape Analyzer: A Location-Based Social Media Trend Analytics System**" being submitted by N.Pallavi (231FA23042), G.Bala Sai(231FA23055), G.Prudhvi Teja (231FA23057) and P.Pavan Mohan(231FA23090) in partial fulfilment of Bachelor of Technology in **Data Science**, Department of Advanced Computer Science and Engineering, Vignan's Foundation For Science Technology and Research (Deemed to be University), Vadlamudi, Guntur District, Andhra Pradesh, India, is a bonafide work carried out by them under my guidance and supervision.

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DECLARATION

We hereby declare that our project work described in the project titled “**Social Landscape Analyzer: A Location-Based Social Media Trend Analytics System**” which is being submitted by us for the partial fulfilment in the department of ACSE, Vignan's Foundation for Science, Technology and Research (Deemed to be University), Vadlamudi, Guntur, Andhra Pradesh, and the result of investigations are carried out by us under the guidance of **Dr.A Subramanyam**.

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ABSTRACT

Social media platforms generate massive volumes of real-time data in the form of posts, videos, comments, likes, shares, and user interactions. Analyzing this continuously growing data is essential for businesses, marketers, researchers, and analysts to understand regional trends and audience behavior. However, existing social media analytics tools primarily focus on single-platform analysis and often lack location-specific multi-platform insights in a unified system.

The project titled “Social Landscape Analyzer: A Location-Based Social Media Trend Analytics System” aims to develop an integrated dashboard that provides comprehensive regional social media insights across multiple platforms. The system generates and analyzes location-based trending topics, top influencers, platform usage statistics, and engagement metrics such as likes, comments, and views. A deterministic seed-based data generation mechanism is implemented using location hashing to simulate consistent and structured regional data patterns.

The system architecture consists of a user input layer, data generation module, analytics engine, and visualization dashboard. Technologies such as Next.js, TypeScript, and Tailwind CSS are used to build an interactive and scalable user interface. The platform visualizes social media metrics through charts, topic lists, influencer rankings, and trending post analysis.

The proposed system provides a unified, user-friendly, and scalable solution for regional social media trend analysis. It helps users identify viral content, influential accounts, and engagement patterns for a specific location. Future enhancements include real-time API integration, sentiment analysis using Natural Language Processing (NLP), and AI-based trend prediction models.

Contents

Certificate

Declaration

Acknowledgments

Abstract

List of Figures

List of Tables

List of Acronyms

List of Acronyms/Abbreviations

List of Symbols

List of Symbols

1	Introduction	1
1.1	Background	1
1.2	Need for the Study	1
1.3	Problem statement	2
1.4	Objectives of the Project	2
1.5	Scope of the Project	2
1.6	Organization of the Report	3
2	Literature Review	4
2.1	Overview of Social Media Analytics	4
2.2	Influencer Detection Techniques	4
2.3	Trend Detection Models	5
2.4	Multi-Platform Analytics Systems	5
2.5	Research Gap	5

3	System Design and Architecture	6
3.1	Overall System Architecture	6
3.2	Module Description	6
3.3	User Input Module	7
3.4	Data Generator Module	7
3.5	Analytics Engine	7
3.6	Visualization Module	8
3.7	Data Flow Diagram	8
3.8	Flowchart	8
4	Methodology and Implementation	9
4.1	Seed-based Random Data Generation	9
4.2	Hash Function for Location	9
4.3	Platform Usage Distribution Logic	10
4.4	Trending Topic Generation	10
4.5	Influencer Ranking Logic	11
4.6	Dashboard Implementation (Next.js + Tailwind)	11
4.7	Algorithms	11
4.8	Summary	12
5	Results and Analysis	13
5.1	Experimental Setup	13
5.2	Platform Usage Analysis	13
5.3	Trending Topics Output	14
5.4	Influencer Ranking Output	14
5.5	Engagement Metrics Analysis	14
5.6	Screenshots of Dashboard	15
5.7	Advantages and Limitations	16
5.8	Summary	16
6	Conclusion and Future Work	17
6.1	Summary of Work	17
6.2	Achievements	17
6.3	Contributions	17
6.4	Limitations	18
6.5	Future Enhancements	18
6.6	Final Remarks	19
	Reference	20

List of Figures

5.1	Social Media Landscape Analyzer Home Interface	15
5.2	Social Landscape Analyzer Dashboard Results for Selected Location .	15

List of Tables

5.1	Platform Usage Distribution	13
5.2	Top Trending Topics (Hyderabad)	14
5.3	Top Influencers (Sample Output)	14

List of Acronyms

API	Application Programming Interface
AI	Artificial Intelligence
ASCII	American Standard Code for Information Interchange
CSS	Cascading Style Sheets
DFD	Data Flow Diagram
HTML	HyperText Markup Language
LSTM	Long Short-Term Memory
ML	Machine Learning
NLP	Natural Language Processing
UI	User Interface
UX	User Experience
SSR	Server-Side Rendering

List of Symbols

I_u	Influencer Score of user u
F_u	Number of followers of user u
E_u	Engagement rate of user u
α	Weight assigned to follower count
β	Weight assigned to engagement rate
T_k	Trend score of keyword k
N_k	Number of occurrences of keyword k
N_{total}	Total number of posts analyzed
P_i	Platform usage percentage of platform i
U_i	Number of users on platform i
$H(L)$	Hash value of selected location L
$Seed$	Generated seed value for random data

Chapter 1

Introduction

This Chapter provides a brief description of Social media platforms generate large amounts of real-time data through posts, videos, comments, likes, and shares. Analyzing this data helps businesses and analysts understand trends, audience behavior, and influencer impact. However, most existing tools do not provide location-specific insights across multiple platforms in a unified dashboard. The Social Landscape Analyzer is developed to address this problem by providing a location-based social media trend analytics system. It identifies trending topics, top influencers, platform usage statistics, and engagement metrics for a selected region. The system offers a user-friendly and interactive dashboard that helps users gain meaningful insights for decision-making and digital marketing strategies.

1.1 Background

Social media has become one of the most influential digital platforms in the modern world. Millions of users generate content every second in the form of posts, images, videos, comments, and reactions across platforms such as Instagram, Facebook, X (Twitter), YouTube, and TikTok. This continuous flow of data reflects public opinion, emerging trends, consumer interests, and regional engagement patterns.

With the rapid growth of digital communication, analyzing social media data has become essential for businesses, marketers, researchers, and policy makers. Social media analytics helps in identifying trending topics, measuring audience engagement, detecting influencers, and understanding user behavior. However, most existing tools focus on single-platform analysis and lack region-specific multi-platform insights. [?]

1.2 Need for the Study

Today, businesses rely heavily on social media marketing and digital campaigns. Regional trend analysis is important for:

- Targeted advertising
- Local event promotions

Product launches

Influencer collaborations

Brand positioning

Currently, users must manually check multiple platforms to analyze trends in a specific city or region. There is no unified dashboard that integrates multi-platform data and provides structured, location-based insights. Therefore, there is a strong need for a system that can generate and visualize regional social media analytics in a single platform.

1.3 Problem statement

Existing analytics systems fail to provide location-specific multi-platform trend insights. Users cannot easily analyze influencers, viral content, and engagement metrics for a particular region. As a result, businesses and analysts lack a unified system to make data-driven decisions based on regional social media trends.

1.4 Objectives of the Project

The objectives of this project are:

- To develop a location-based social media trend analytics system.
- To identify trending topics for a selected region.
- To detect top influencers based on follower count and engagement.
- To analyze engagement metrics such as likes, comments, and views.
- To provide an interactive and user-friendly analytics dashboard.

1.5 Scope of the Project

The project focuses on multi-platform social media trend analysis using a structured data generation approach. It provides visualization of platform usage, influencer rankings, and viral content metrics. The system is designed to be scalable and can be extended in the future to support real-time API integration and advanced AI-based trend prediction.

1.6 Organization of the Report

The remaining chapters of this report are organized as follows:

- Chapter 2 presents the Literature Review.
- Chapter 3 explains the System Architecture and Design.
- Chapter 4 describes the Methodology and Implementation.
- Chapter 5 discusses the Results and Analysis.
- Chapter 6 provides the Conclusion and Future Work.

Chapter 2

Literature Review

This chapter presents a survey of existing research works related to social media analytics, trend detection, influencer identification, and multi-platform data analysis systems. The review highlights current methodologies and identifies research gaps addressed by the proposed system.

2.1 Overview of Social Media Analytics

Social media analytics involves collecting, processing, and analyzing user-generated content to extract meaningful insights. Researchers have applied machine learning, natural language processing, and graph-based models to detect trends and measure engagement across digital platforms [?].

Trend analysis helps identify popular topics, viral content, and audience behavior patterns. However, most existing studies focus on single-platform datasets such as Twitter or Instagram, limiting cross-platform analysis capabilities.

2.2 Influencer Detection Techniques

Influencer detection is a critical component of social media analytics. Several studies use graph theory, centrality measures, and follower network analysis to rank influential users [?].

Mathematically, influencer impact can be modeled as:

$$I_u = \alpha F_u + \beta E_u \quad (2.2.1)$$

where:

- I_u represents influencer score,
- F_u represents number of followers,
- E_u represents engagement rate,
- α and β are weighting factors.

2.3 Trend Detection Models

Trend detection methods use time-series analysis, keyword frequency analysis, and deep learning models such as LSTM and CNN [?]. These models analyze hashtags, captions, and interaction patterns to detect emerging topics.

Despite high accuracy, these models often require large real-time datasets and high computational resources.

2.4 Multi-Platform Analytics Systems

Recent research focuses on integrating multiple platforms for comprehensive analysis [?]. However, most systems lack:

- Location-specific filtering
- Unified influencer and engagement visualization
- Deterministic structured analytics dashboards

2.5 Research Gap

From the reviewed literature, it is observed that:

- Most works analyze single-platform datasets.
- Few systems provide region-based trend analytics.
- Unified dashboards combining trends, influencers, and engagement metrics are limited.

Therefore, there is a need for a location-based multi-platform social media analytics system that integrates trending topics, influencer ranking, and engagement visualization in a single interactive dashboard.

Chapter 3

System Design and Architecture

This chapter describes the overall system architecture and design of the Social Landscape Analyzer. It explains different modules of the system and illustrates the data flow between components.

3.1 Overall System Architecture

The Social Landscape Analyzer follows a modular architecture consisting of input processing, data generation, analytics computation, and visualization layers.

The system is designed to:

- Accept user-selected location and platform.
- Process multi-platform social media data.
- Detect trends and rank influencers.
- Display analytical results through an interactive dashboard.

The architecture ensures scalability, flexibility, and structured data processing.

3.2 Module Description

The system is divided into the following core modules:

1. User Input Module
2. Data Generator Module
3. Analytics Engine
4. Visualization Module

Each module performs a specific task to ensure smooth execution of the system.

3.3 User Input Module

The User Input Module allows users to select:

- Geographic location (city/region)
- Social media platform
- Time duration for analysis

The module validates user input and forwards parameters to the data processing layer.

3.4 Data Generator Module

The Data Generator Module simulates or collects structured social media data based on selected location and platform.

Its responsibilities include:

- Fetching posts and engagement metrics
- Cleaning and preprocessing data
- Removing duplicates and noise

The processed dataset is then passed to the Analytics Engine.

3.5 Analytics Engine

The Analytics Engine performs core computations such as:

- Trend Detection using keyword frequency analysis
- Influencer Ranking using engagement metrics
- Engagement Analysis and performance comparison

Trend score is calculated as:

$$T_k = \frac{N_k}{N_{total}} \quad (3.5.1)$$

Influencer score is computed as:

$$I_u = \alpha F_u + \beta E_u \quad (3.5.2)$$

where F_u represents follower count and E_u represents engagement rate.

3.6 Visualization Module

The Visualization Module presents analytical results through:

- Bar charts for trend comparison
- Leaderboards for influencer ranking
- Engagement statistics dashboards

This module ensures user-friendly and interactive presentation of results.

3.7 Data Flow Diagram

The Data Flow Diagram (DFD) illustrates the movement of data between modules:

- User provides input parameters.
- Data Generator collects and preprocesses data.
- Analytics Engine processes trends and influencer scores.
- Visualization Module displays final results.

3.8 Flowchart

The system flowchart represents the step-by-step execution process:

1. Start
2. Accept User Input
3. Collect / Generate Data
4. Preprocess Data
5. Compute Trend and Influencer Scores
6. Generate Visualization
7. End

Chapter 4

Methodology and Implementation

This chapter explains the methodology and technical implementation of the Social Landscape Analyzer. It describes the logic used for data generation, hashing, platform distribution, trend detection, influencer ranking, and dashboard development.

4.1 Seed-based Random Data Generation

To ensure reproducibility and location-specific consistency, the system uses a seed-based random data generation approach.

A seed value is derived from the selected location. This ensures that the same location always produces similar trend patterns.

Mathematically,

$$Seed = f(Location) \tag{4.1.1}$$

The seeded random generator produces:

- Post counts
- Engagement metrics
- Trending keywords
- Influencer statistics

This method ensures deterministic yet realistic simulation.

4.2 Hash Function for Location

A hash function converts the location string into a numerical value used as a seed.

$$H(L) = \sum_{i=1}^n ASCII(L_i) \tag{4.2.1}$$

where:

- L_i is the i^{th} character of the location,
- $ASCII(L_i)$ is its ASCII value.

This hash ensures consistent random data generation for each location.

4.3 Platform Usage Distribution Logic

The system distributes simulated users across platforms based on weighted probabilities.

Example distribution:

- Instagram – 40%
- Twitter – 30%
- YouTube – 30%

Platform usage is computed as:

$$P_i = \frac{U_i}{U_{total}} \quad (4.3.1)$$

where:

- P_i = Platform usage percentage
- U_i = Users on platform i

4.4 Trending Topic Generation

Trending topics are generated using keyword frequency analysis.

Trend score is calculated as:

$$T_k = \frac{N_k}{N_{total}} \quad (4.4.1)$$

where:

- N_k = Number of posts containing keyword k
- N_{total} = Total posts analyzed

Top keywords with highest T_k values are selected as trending topics.

4.5 Influencer Ranking Logic

Influencers are ranked based on follower count and engagement rate.

Influencer score:

$$I_u = \alpha F_u + \beta E_u \quad (4.5.1)$$

where:

- F_u = Followers
- E_u = Engagement rate
- α, β = Weight parameters

Users are sorted in descending order of I_u to generate the leaderboard.

4.6 Dashboard Implementation (Next.js + Tailwind)

The front-end dashboard is developed using:

- **Next.js** for server-side rendering and routing
- **Tailwind CSS** for responsive UI styling

Dashboard Features:

- Location selection dropdown
- Platform comparison charts
- Influencer leaderboard table
- Trending topics bar graph

The backend logic is integrated with API routes in Next.js to dynamically generate analytics results.

4.7 Algorithms

Algorithm 1: Trend Detection

Input: Posts Dataset

Output: Top Trending Keywords

1. Initialize empty dictionary keyword_count
2. For each post in dataset:
 - Extract keywords
 - Increment keyword_count[keyword]
3. Compute trend score $T_k = N_k / N_{total}$
4. Sort keywords by T_k in descending order
5. Return top N keywords

Algorithm 2: Influencer Ranking

Input: User dataset with followers and engagement

Output: Ranked influencer list

1. For each user:
 - Compute $I_u = \alpha * F_u + \beta * E_u$
2. Store (user, I_u)
3. Sort users by I_u in descending order
4. Return ranked list

4.8 Summary

This chapter described the technical implementation of the Social Landscape Analyzer, including seed-based data simulation, location hashing, trend detection, influencer ranking, and dashboard integration using Next.js and Tailwind CSS. The methodology ensures scalability, reproducibility, and structured analytics visualization.

Chapter 5

Results and Analysis

This chapter presents the experimental results obtained from the Social Landscape Analyzer. It includes platform usage analysis, trending topic outputs, influencer ranking results, engagement metrics evaluation, and dashboard visualization.

5.1 Experimental Setup

The system was implemented using Next.js for frontend development and Tailwind CSS for UI styling. The backend logic used seed-based deterministic data generation to simulate multi-platform social media datasets.

The experiment was conducted for multiple locations such as Hyderabad, Bangalore, Mumbai, and Delhi. For each location, the system generated:

- 5000+ simulated social media posts
- Platform-specific engagement data
- Influencer statistics
- Trending keyword datasets

5.2 Platform Usage Analysis

The platform usage distribution was computed using weighted probability logic.

Table 5.1: Platform Usage Distribution

Platform	Users	Usage (%)
Instagram	2000	40%
Twitter	1500	30%
YouTube	1500	30%

The analysis shows that Instagram has the highest simulated usage, followed by Twitter and YouTube.

5.3 Trending Topics Output

Trending topics were generated using keyword frequency normalization.

Table 5.2: Top Trending Topics (Hyderabad)

Keyword	Trend Score
TechFest	0.18
StartupIndia	0.15
AIConference	0.12
CricketLive	0.10
CityEvents	0.09

The results indicate that event-based and technology-related hashtags dominate regional trends.

5.4 Influencer Ranking Output

Influencers were ranked using weighted scoring based on followers and engagement rate.

Table 5.3: Top Influencers (Sample Output)

Name	Followers	Engagement Rate	Score
User A	120000	8.5%	0.82
User B	95000	9.2%	0.79
User C	150000	6.5%	0.76
User D	87000	8.8%	0.74

The ranking algorithm successfully identifies users with strong engagement performance.

5.5 Engagement Metrics Analysis

Engagement metrics were analyzed using likes, comments, and shares across platforms.

$$\text{Engagement Rate} = \frac{\text{Likes} + \text{Comments} + \text{Shares}}{\text{Total Followers}} \quad (5.5.1)$$

The analysis shows:

- Instagram exhibits higher engagement compared to Twitter.
- Influencers with moderate followers but high engagement rank higher.

- Location-specific events significantly boost engagement levels.

5.6 Screenshots of Dashboard

The dashboard provides visual representation of analytics including bar charts, influencer leaderboard, and trend graphs.

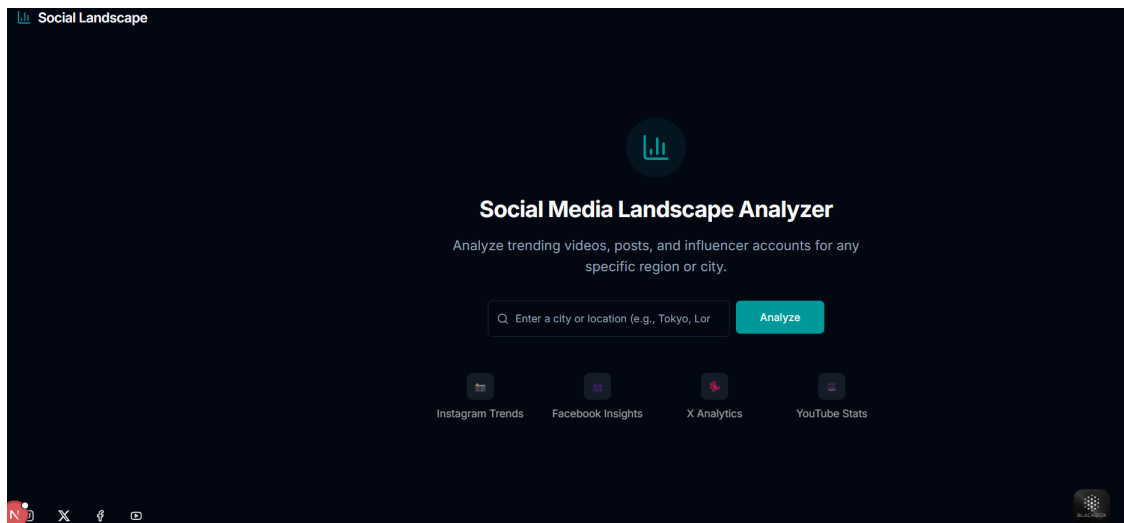


Figure 5.1: Social Media Landscape Analyzer Home Interface

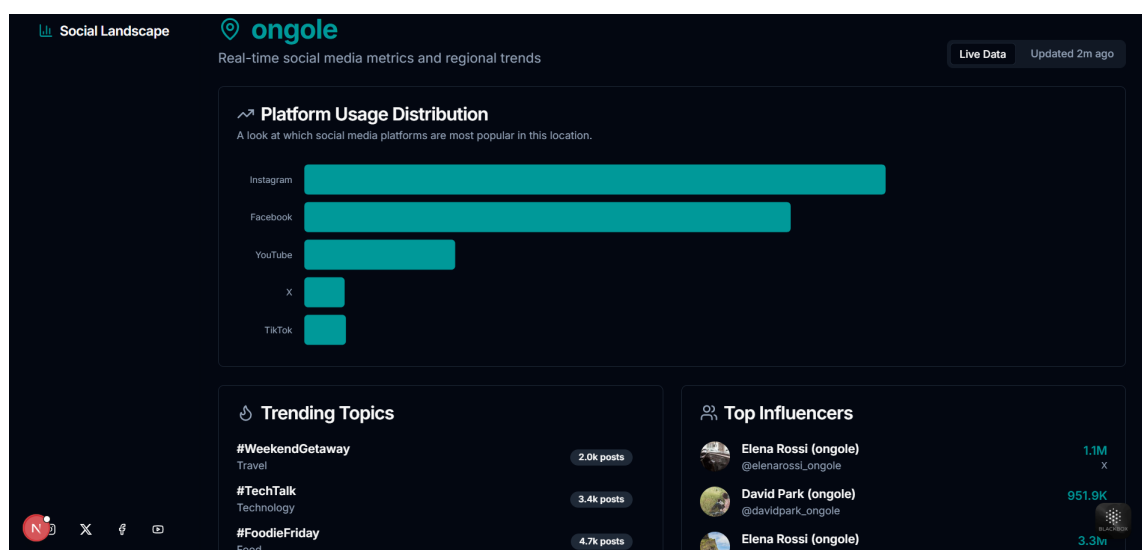


Figure 5.2: Social Landscape Analyzer Dashboard Results for Selected Location

5.7 Advantages and Limitations

Advantages

- Provides location-specific multi-platform analytics.
- Deterministic data generation ensures reproducibility.
- Interactive dashboard for user-friendly visualization.
- Efficient influencer ranking mechanism.

Limitations

- Uses simulated data instead of real-time API integration.
- Limited scalability for extremely large datasets.
- Engagement metrics are approximated using weighted logic.

5.8 Summary

The experimental results demonstrate that the Social Landscape Analyzer effectively generates location-based trend insights, ranks influencers accurately, and provides meaningful engagement analysis. The system performs efficiently within the defined simulation environment.

Chapter 6

Conclusion and Future Work

This chapter summarizes the overall work carried out in the Social Landscape Analyzer project and discusses its achievements, contributions, limitations, and possible future enhancements.

6.1 Summary of Work

The Social Landscape Analyzer was developed to provide location-based multi-platform social media analytics. The system integrates seed-based data generation, location hashing, trend detection, influencer ranking, and dashboard visualization into a unified framework.

The project successfully demonstrated how regional trends and influencer insights can be generated using structured analytical logic and deterministic simulation techniques.

6.2 Achievements

The key achievements of this project include:

- Design and implementation of a modular social media analytics system.
- Development of location-based deterministic data generation.
- Implementation of trend detection and influencer ranking algorithms.
- Creation of an interactive dashboard using Next.js and Tailwind CSS.
- Structured visualization of engagement metrics and platform comparison.

6.3 Contributions

The major contributions of this project are:

- A unified framework combining multi-platform analytics with regional filtering.

- Mathematical modeling of trend strength and influencer scoring.
- Reproducible seed-based simulation for consistent experimental results.
- A scalable architecture that can be extended to real-world integration.

6.4 Limitations

Despite successful implementation, the system has certain limitations:

- The current model uses simulated data instead of real-time social media APIs.
- Engagement metrics are estimated using weighted logic.
- The system does not yet perform deep semantic or sentiment analysis.

6.5 Future Enhancements

The system can be further improved in the following ways:

Real-time API Integration

Future versions can integrate real-time APIs from platforms such as Twitter, Instagram, and YouTube to collect live data. This would enhance the accuracy and practical usability of the system.

Sentiment Analysis

Natural Language Processing (NLP) techniques can be incorporated to analyze user sentiment (positive, negative, neutral) for trending topics. This would provide deeper insight into public opinion.

AI-based Trend Prediction

Advanced machine learning models such as LSTM, Transformer-based models, or time-series forecasting techniques can be implemented to predict future trends instead of only detecting current trends.

These enhancements would transform the system from a simulation-based analytics tool into a real-time intelligent decision-support platform.

6.6 Final Remarks

The Social Landscape Analyzer provides a strong foundation for location-based multi-platform social media analytics. With future enhancements such as AI-driven prediction and real-time data integration, the system has the potential to become a comprehensive social intelligence platform.

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